

# Surface Area of Prisms

Lesson 10-4

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

## Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*Paint every outside surface of a box — the total painted area is the surface area*

**Surface area**

The total area of all the flat sides of a solid.

*A box with 6 flat rectangle sides: top, bottom, front, back, left, right*

**Rectangular prism**

A solid box shape with six flat rectangle sides.

*Shaped like a tent or a wedge of cheese — 2 triangle ends + 3 rectangle sides*

**Triangular prism**

A solid with two triangle ends and three flat rectangle sides.

*A triangular prism has 5 faces: 2 triangles + 3 rectangles*

**Face**

One flat side of a solid shape.

*Unfold a triangular prism flat: you see 2 triangles and 3 rectangles side by side*

**Net**

A flat shape that folds up into a solid.

*On a triangular prism, the 3 rectangles that wrap around the sides are lateral faces*

**Lateral face**

The side faces of a solid, not the top or bottom.

## Key Ideas & Notes



### WHAT WE'RE LEARNING TODAY

**I can find the surface area of rectangular and triangular prisms.**

#### 1 Notice

Your class is painting the outside of different time capsule shapes to protect them from weather.

#### 2 Key idea

I can find the surface area of rectangular and triangular prisms.

#### 3 Words I'll use

Surface area

Rectangular prism

Triangular prism

Face

Net

Lateral face



#### Think About It

- How many faces does a rectangular prism have? A triangular prism?
- What shapes make up the faces of a triangular prism?
- How is finding surface area different from finding volume?



#### Watch

Watch your teacher model one example. Jot what you see.

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#### We try

Solve the next one together as a class.

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#### You try

Now try one on your own.

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## Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

### LAUNCH

**Your class is painting two capsule shapes: a rectangular prism and a triangular prism. How is finding the surface area of each shape the same, and how is it different?**

#### Level 1 support

**Start here:** For both, you add the areas of all the faces, but the shapes and number of faces differ.

#### 💡 Need a hint?

**Sentence starters (optional)**

**Both surface areas are found by adding the areas of all the \_\_\_\_.**

*Ambas áreas de superficie se hallan sumando las áreas de todas las \_\_\_\_.*

**A rectangular prism has \_\_\_\_ faces, but a triangular prism has \_\_\_\_ faces.**

*Un prisma rectangular tiene \_\_\_\_ caras, pero un prisma triangular tiene \_\_\_\_ caras.*

**Word bank:** surface area rectangular prism triangular prism face lateral face

**Listen for:** Students explain both methods add up all face areas, but a rectangular prism has 6 rectangular faces while a triangular prism has 5 faces (2 triangles + 3 rectangles).

#### Level 2

**Why does surface area use square units while volume uses cubic units? Justify with what each measurement counts.**

- Surface area uses square units because it measures \_\_\_\_.
- Volume uses cubic units because it measures \_\_\_\_.

**EXPLORE**

**For the triangular prism (base 6, height 4, length 10, slant sides 5), how did you account for the triangular bases AND the rectangular lateral faces?**

**Level 1 support**

**Start here:** Add the two triangular bases to the three rectangular lateral faces.



**Need a hint?**

**Sentence starters (optional)**

**The two triangular bases together are  $2 \times (\frac{1}{2} \times 6 \times 4) = \underline{\hspace{1cm}}$ .**

*Las dos bases triangulares juntas son  $2 \times (\frac{1}{2} \times 6 \times 4) = \underline{\hspace{1cm}}$ .*

**The three lateral faces add to  $\underline{\hspace{1cm}}$ , so the total surface area is  $\underline{\hspace{1cm}}$ .**

*Las tres caras laterales suman  $\underline{\hspace{1cm}}$ , así que el área total de superficie es  $\underline{\hspace{1cm}}$ .*

**Word bank:** **surface area** **triangular prism** **lateral face** **face** **net**

**Listen for:** Students compute the two triangle bases ( $24 \text{ in}^2$ ) and the three rectangles ( $60 + 50 + 50 = 160 \text{ in}^2$ ), totaling  $SA = 184 \text{ in}^2$ , and distinguish bases from lateral faces.

**Level 2**

**Compare the rectangular capsule ( $SA = 248 \text{ in}^2$ ) with the similar triangular capsule ( $SA = 184 \text{ in}^2$ ). Why does the rectangular one need more paint?**

- The rectangular prism needs more paint because it has  $\underline{\hspace{1cm}}$  faces and  $\underline{\hspace{1cm}}$ .
- Even at a similar size, the triangular prism uses less because  $\underline{\hspace{1cm}}$ .

**CONNECT**

A carpenter stains a chest that is  $3\text{ ft} \times 2\text{ ft} \times 2\text{ ft}$ , and one can covers  $50\text{ ft}^2$ . Talk through how to find the surface area and how many cans are needed.

**Level 1 support**

**Start here:** Find the surface area with  $SA = 2lw + 2lh + 2wh$ , then divide by 50 and round up.



**Need a hint?**

**Sentence starters (optional)**

**The chest's surface area is \_\_\_\_  $\text{ft}^2$  because  $SA = 2(3 \times 2) + 2(3 \times 2) + 2(2 \times 2)$ .**

*El área de superficie del cofre es \_\_\_\_  $\text{ft}^2$  porque  $SA = 2(3 \times 2) + 2(3 \times 2) + 2(2 \times 2)$ .*

**The carpenter needs \_\_\_\_ can(s) because \_\_\_\_  $\div 50 =$  \_\_\_\_.**

*El carpintero necesita \_\_\_\_ lata(s) porque \_\_\_\_  $\div 50 =$  \_\_\_\_.*

**Word bank:** **surface area** **rectangular prism** **face** **lateral face** **net**

**Listen for:** Students compute  $SA = 12 + 12 + 8 = 32\text{ ft}^2$ , then  $32 \div 50 < 1$ , and reason that 1 can is enough since the surface area is under  $50\text{ ft}^2$ .

**Level 2**

**If the chest had a lid that does not get stained, how would that change the surface area you calculate? Explain.**

- Leaving the lid unstained changes the surface area by \_\_\_\_ because \_\_\_\_.
- I would subtract \_\_\_\_ from the total because \_\_\_\_.

**REFLECT**

A classmate wrote  $SA = 2(12 \times 7) + 2(12 \times 5) + (7 \times 5) = 323 \text{ cm}^2$  for a  $12 \times 7 \times 5$  prism. How do you find and fix their mistake?

**Level 1 support**

**Start here:** They forgot to multiply the last face pair by 2; the correct SA is  $358 \text{ cm}^2$ .

 **Need a hint?****Sentence starters (optional)**

**The mistake is that the last pair ( $7 \times 5$ ) was not multiplied by \_\_\_\_.**

*El error es que el último par ( $7 \times 5$ ) no se multiplicó por \_\_\_\_.*

**The correct surface area is \_\_\_\_  $\text{cm}^2$  because  $2(7 \times 5) =$  \_\_\_\_, not \_\_\_\_.**

*El área de superficie correcta es \_\_\_\_  $\text{cm}^2$  porque  $2(7 \times 5) =$  \_\_\_\_, no \_\_\_\_.*

**Word bank:** **surface area** **rectangular prism** **face** **lateral face** **net**

**Listen for:** Students notice the missing '2' on the last pair, compute  $2(7 \times 5) = 70$ , and correct the total to  $168 + 120 + 70 = 358 \text{ cm}^2$ .

**Level 2**

**What general rule about prism faces does this error break, and how could the net help a student avoid it?**

- The error breaks the rule that opposite faces \_\_\_\_.
- Drawing the net would help because \_\_\_\_.

## Guided Examples

### Example 1

**What is the surface area of a rectangular prism with  $l = 5$  cm,  $w = 4$  cm,  $h = 3$  cm?**

**Solution:**  $SA = 2(5 \times 4) + 2(5 \times 3) + 2(4 \times 3) = 40 + 30 + 24 = 94 \text{ cm}^2$ .

**Answer:** A.  $94 \text{ cm}^2$

### Example 2

**How many faces does a triangular prism have?**

**Solution:** A triangular prism has 5 faces: 2 triangular bases and 3 rectangular lateral faces.

**Answer:** A. 5

### Example 3

**A triangular prism has two triangular bases each with area  $12 \text{ cm}^2$  and three rectangular faces with areas  $30 \text{ cm}^2$ ,  $40 \text{ cm}^2$ , and  $40 \text{ cm}^2$ . What is the total surface area?**

**Solution:**  $SA = 2(12) + 30 + 40 + 40 = 24 + 110 = 134 \text{ cm}^2$ .

**Answer:** A.  $134 \text{ cm}^2$

## Write About the Math

### The Writing Revolution

I can explain my work using the words surface area, rectangular prism, triangular prism, and face.

#### 1. Kernel Sentence subject + verb

**Model:** Surface area is the total area of all the flat sides of a solid.

*Área de superficie es el área total de todos los lados planos de un sólido.*

**Write a kernel sentence about surface area. Use a subject and a verb.**

*Escribe una oración base sobre área de superficie. Usa un sujeto y un verbo.*

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#### 2. Sentence Expansion because · but · so

**Kernel:** Surface area matters in math

*Área de superficie importa en matemáticas*

Expand the kernel three ways. Add a reason, a contrast, and a result.

**because**  
*porque*

**Surface area matters in math because \_\_\_\_.**

*Área de superficie importa en matemáticas porque \_\_\_\_.*

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**but**  
*pero*

**Surface area matters in math, but \_\_\_\_.**

*Área de superficie importa en matemáticas, pero \_\_\_\_.*

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**so**  
*entonces*

**Surface area matters in math, so \_\_\_\_.**

*Área de superficie importa en matemáticas, entonces \_\_\_\_.*

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### 3. Sentence Types 4 ways to write a math idea

**Statement**  
*Afirmación*

Tell one true fact about surface area.  
*Di un hecho verdadero sobre surface area.*

**Surface area** \_\_\_\_.

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**Question**  
*Pregunta*

Ask a question about surface area.  
*Haz una pregunta sobre surface area.*

**How does** \_\_\_\_ ?

*¿Cómo* \_\_\_\_ ?

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**Exclamation**  
*Exclamación*

Show excitement about surface area.  
*Muestra entusiasmo sobre surface area.*

**Wow,** \_\_\_\_ !

*¡Guau,* \_\_\_\_ !

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**Command**  
*Mandato*

Tell a partner what to do with surface area.  
*Dile a un compañero qué hacer con surface area.*

**First,** \_\_\_\_ .

*Primero,* \_\_\_\_ .

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### 4. Explain Your Reasoning use a sentence starter

**Both surface areas are found by adding the areas of all the** \_\_\_\_ .

*Ambas áreas de superficie se hallan sumando las áreas de todas las* \_\_\_\_ .

**A rectangular prism has** \_\_\_\_ **faces, but a triangular prism has** \_\_\_\_ **faces.**

*Un prisma rectangular tiene* \_\_\_\_ *caras, pero un prisma triangular tiene* \_\_\_\_ *caras.*

**The two triangular bases together are**  $2 \times (\frac{1}{2} \times 6 \times 4) =$  \_\_\_\_ .

*Las dos bases triangulares juntas son*  $2 \times (\frac{1}{2} \times 6 \times 4) =$  \_\_\_\_ .

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## Try It

Solve on your own. Check the answer key when you are done.

**1. A triangular prism has two triangular bases each with area  $12 \text{ cm}^2$  and three rectangular faces with areas  $30 \text{ cm}^2$ ,  $40 \text{ cm}^2$ , and  $40 \text{ cm}^2$ . What is the total surface area?**

A.  $134 \text{ cm}^2$

B.  $110 \text{ cm}^2$

C.  $122 \text{ cm}^2$

D.  $134 \text{ cm}^3$

Show your work:

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**2. A triangular prism has triangular bases with base 6 in and height 4 in. The prism is 10 in long and the two slant sides of the triangle are each 5 in. What is the surface area?**

A.  $184 \text{ in}^2$

B.  $160 \text{ in}^2$

C.  $124 \text{ in}^2$

D.  $184 \text{ in}^3$

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**A company ships products in two container options: a rectangular prism ( $8 \times 6 \times 4$  in) or a triangular prism (triangle base 8 in, triangle height 6 in, prism length 8 in, slant sides 6.3 in each). Both must be wrapped in protective film. Which shape uses LESS film? Show your work.**

*Sentence starter: Rectangular prism  $SA = 2(\text{ }) + 2(\text{ }) + 2(\text{ }) = \text{ } \text{ in}^2$ . Triangular prism  $SA = 2(\text{ }) + \text{ } + \text{ } + \text{ } = \text{ } \text{ in}^2$ . The  $\text{ }$  uses less film because  $\text{ }$ .*

Show your work:

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## Reflect — Exit Ticket

**A rectangular prism has  $l = 8$  ft,  $w = 6$  ft,  $h = 3$  ft. What is the surface area?**

- A.  $180 \text{ ft}^2$
- B.  $144 \text{ ft}^2$
- C.  $180 \text{ ft}^3$
- D.  $90 \text{ ft}^2$

Your answer:

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## Answer Key & Teacher Guide

1. **Try It 1:** A.  $134 \text{ cm}^2$  —  $SA = 2(12) + 30 + 40 + 40 = 24 + 110 = 134 \text{ cm}^2$ .
2. **Try It 2:** A.  $184 \text{ in}^2$  — *Triangular bases:*  $2 \times (\frac{1}{2} \times 6 \times 4) = 24 \text{ in}^2$ . *Three rectangular faces:*  $6(10) + 5(10) + 5(10) = 60 + 50 + 50 = 160 \text{ in}^2$ .  $SA = 24 + 160 = 184 \text{ in}^2$ .
3. **Exit Ticket:** A.  $180 \text{ ft}^2$  —  $SA = 2(8 \times 6) + 2(8 \times 3) + 2(6 \times 3) = 96 + 48 + 36 = 180 \text{ ft}^2$ . *Surface area uses square units ( $\text{ft}^2$ ).*

### Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Surface area is the total area of all the flat sides of a solid.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).